

Chapter 5 – Thermochemistry

5.1. Energy

1. Understand **work** as a force displacing an object and recognize it as a way that energy can be transferred to or from an object.
2. Understand **heat** as energy transferred from one object to another driven by a temperature difference between the objects.
3. Understand **kinetic energy (KE)** as energy of motion. Know the equation, $KE = \frac{1}{2}mv^2$, and be able to calculate kinetic energy using it.
4. Understand **potential energy (PE)** as energy of position and recognize that **chemical energy** is the potential energy arising from the atom positions within a molecule.
5. Know that the SI energy unit is the **joule (J)** which is equal to $1 \text{ kg}\cdot\text{m}^2/\text{s}^2$ and be able to convert between different energy units given the conversion factor.

5.2. Enthalpy

1. Understand **enthalpy** as a thermodynamic function represented by H and given by the sum of the internal energy (E) and the product of pressure (P) and volume (V). Mathematically, $H = E + PV$.
2. Know the physical interpretation of an **enthalpy change (ΔH)** as the heat transferred at constant pressure.
3. Know that in an **exothermic process**, heat is transferred from the system to the surroundings and $\Delta H < 0$.
4. Know that in an **endothermic process**, heat is transferred from the surroundings to the system and $\Delta H > 0$.

5.3. Energy, Temperature Changes, and Changes of State

1. Be able to define **specific heat capacity** and be able to determine the specific heat capacity of a substance from the temperature change occurring with a specified amount of heat transferred into or out of a specified mass of the substance.
2. Be able to use $q = m \times C \times \Delta T$ in heat transfer calculations.
3. Be able to define **thermal equilibrium**.
4. Be able to calculate the heat transferred for a phase transition of a specified amount of a substance.
5. Be able to identify and interpret the components of a **heating curve**.
6. Be able to calculate the heat transferred in every component of a heating curve.

5.4. Enthalpy Changes and Chemical Reactions

1. Know that the **enthalpy of reaction (ΔH_{rxn})** is the enthalpy change for the chemical conversion of reactants into products in the mole amounts shown in the balanced chemical reaction.
2. Recognize that reversing the direction of a reaction changes the sign of ΔH .

3. Recognize that multiplying a reaction by a constant multiplies ΔH by the same constant.
 4. Recognize that for an **exothermic reaction**
 - a. $\Delta H_{\text{rxn}} < 0$
 - b. can be represented as: reactants $\rightarrow$ products $- \Delta H_{\text{rxn}}$
 - c. enthalpy of reactants is *greater* than enthalpy of products
 - d. energy in the form of heat is *released* as reactants are converted into products
 - e. temperature will *increase*.
 5. Recognize that for an **endothermic reaction**
 - a. $\Delta H_{\text{rxn}} > 0$
 - b. can be represented as: reactants $+ \Delta H_{\text{rxn}} \rightarrow$ products
 - c. enthalpy of reactants is *less* than enthalpy of products
 - d. energy in the form of heat is *absorbed* as reactants are converted into products
 - e. temperature will *decrease*.
 6. Be able to define a **calorimeter** and describe its role in **calorimetry**.
 7. Be able to analyze **constant pressure calorimetry** data like what is obtained using a **coffee-cup calorimeter**.
 8. Be able to analyze **constant volume calorimetry** data like what is obtained using a **bomb calorimeter**.
- 5.5. Hess's Law
1. Be able to define a **state function** and **Hess's law** and explain the role of state functions in the application of Hess's law.
 2. Be able to determine ΔH_{rxn} for a specified reaction of interest using Hess's law and a set of reactions with known ΔH_{rxn} which can be manipulated to sum to the reaction of interest.
- 5.6. Standard Heats of Reaction
1. Be able to define a **standard state**, a **standard state of formation reaction**, and a **standard enthalpy of formation ($\Delta_f H^\circ$)**.
 2. Be able to use standard enthalpy of formation ($\Delta_f H^\circ$) to determine ΔH_{rxn}

Chapter 6 – Electromagnetic Radiation and Atomic Electronic Structure

- 6.1. Electromagnetic Radiation
 1. For an **electromagnetic wave**, be able to define the **wavelength**, **frequency**, and **amplitude**, and know the standard symbol used for wavelength and frequency.
 2. Know the standard symbol used for the **speed of light** and that the wavelength times the frequency equals the speed of light ($\lambda \times \nu = c$).
 3. Know the named regions of the electromagnetic spectrum in wavelength order, **Gamma ray**, **X-ray**, **ultraviolet**, **visible**, **infrared**, **microwave**, and **radio**, and know that the visible region is 400 nm to 700 nm (blue to red).
- 6.2. Photons and Photon Energy

1. Know what is meant by the **photoelectric effect** and that a **photon** is a particle of electromagnetic radiation having energy proportional to its frequency with **Planck's constant** being the proportionality constant.
2. Be able to use $E_{\text{photon}} = h\nu = hc/\lambda$ to determine the energy of a photon from its frequency or wavelength and a photon wavelength and frequency from its energy.

6.3. Atomic Line Spectra and the Bohr Model of Atomic Structure

1. Be able to identify and distinguish **continuous spectra** and **line spectra**.
2. Understand the meaning of **ground state** and **excited state** as these terms are applied to atoms and how the **absorption** and **emission** of a photon allows the interconversion of these states.
3. Understand the role of the **principal quantum number**, n , in determining the allowed energy levels of a hydrogen atom.
4. Be able to use $E_n = -hcR_H \left(\frac{1}{n^2} \right) = -2.1787 \times 10^{-18} \text{ J} \left(\frac{1}{n^2} \right)$ to determine the energy of a hydrogen atom level given the value of n .
5. Be able to use $\Delta E = -2.1787 \times 10^{-18} \text{ J} \left(\frac{1}{n_{\text{final}}^2} - \frac{1}{n_{\text{initial}}^2} \right)$ to determine the energy change for a hydrogen atom given the value for n_{final} and n_{initial} . From the value of the energy change be able to determine the frequency and wavelength of the photon involved in the transition.

6.4. Quantum Theory of Atomic Structure

1. Understand the **de Broglie wavelength** as a manifestation of the wave-particle duality concept, and be able to use $\lambda_B = h/mv$ to determine the de Broglie wavelength or the mass or the velocity of a particle given the value of two of these quantities.
2. Understand the **uncertainty principle** as a consequence of wave-particle duality and be able to use $(\Delta x) \times m(\Delta v) \geq h/4\pi$ to determine Δx or Δv given the value of one of these and m .
3. Understand **orbitals** as three-dimensional functions arising from the **wave functions** which are solution of the **Schrödinger equation** and that orbitals give the probability density of where an electron in that orbital might be found.

6.5. Quantum Numbers and Orbitals

1. Know the allowed values of the **principle quantum number**, **angular momentum quantum number**, and **magnetic quantum number** and what characteristics of the associated orbital are controlled by each quantum number.
2. Know the letters (s, p, d, f) used to label orbitals with values of ℓ value from 0 to 3 and how **subshells** are labeled by the value of n and the letter representation for ℓ .
3. Know the number and general shape of orbitals in a specified subshell.
4. Be able to define and recognize a **node** in a wave function.
5. Be able to identify the n and ℓ value of an orbital based on its node pattern.